

LIARVATN, Norway

WMO: 014190

Lat: 59.0512N

Lon: 6.1213E

Elev: 300

StdP: 97.77

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.5	-9.9	-17.2	0.9	-9.4	-14.7	1.1	-6.9	10.5	-1.5	9.4	-1.3	1.7	20	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	23.9	15.2	21.5	14.2	19.6	13.8	16.5	21.1	15.6	19.6	14.7	18.4	2.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	10.8	17.7	13.8	10.2	17.1	13.0	9.7	16.1	47.2	21.0	44.5	19.6	42.0	18.6	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.0	7.6	6.5		-13.2	26.8	3.1	2.5	-15.4	28.7	-17.2	30.1	-19.0	31.6	-21.2	33.4	
(5)			DB	-13.7	17.7	3.1	1.3	-15.9	18.6	-17.7	19.4	-19.4	20.2	-21.7	21.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-0.3	-0.2	1.6	5.2	9.2	11.8	14.2	13.5	11.3	7.3	3.7	1.1
	DBStd	6.18	3.69	3.95	3.25	3.42	4.13	2.50	3.24	2.34	2.66	2.74	3.57	4.40
	HDD10.0	1665	319	284	259	150	67	9	1	1	15	91	191	277
	HDD18.3	4303	577	518	518	394	284	197	137	150	211	342	441	536
	CDD10.0	414	0	0	0	6	42	62	131	111	55	7	1	0
	CDD18.3	11	0	0	0	0	1	0	8	1	1	0	0	0
	CDH23.3	82	0	0	0	0	13	4	58	4	2	0	0	0
	CDH26.7	8	0	0	0	0	1	0	7	0	0	0	0	0
(14) Wind	WSAvg	2.8	3.3	3.0	2.9	2.8	2.7	2.5	2.3	2.4	2.7	3.0	2.9	3.1
(15) Precipitation	PrecAvg	1764	186	154	138	93	81	95	116	147	176	203	193	198
	PrecMax	2320	376	376	306	158	203	165	260	268	320	407	373	371
	PrecMin	1205	0	0	6	40	4	19	50	36	54	53	22	18
	PrecStd	288	101	98	71	33	46	46	51	54	79	82	102	110
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	8.8	11.9	18.1	25.2	23.5	27.4	23.6	22.0	17.0	12.5	9.4
		MCWB	5.4	5.1	7.7	10.5	13.2	14.4	16.9	15.0	15.4	12.0	9.7	8.1
	2%	DB	6.2	6.4	9.1	16.1	21.9	20.3	25.0	20.7	18.6	14.5	10.8	8.1
		MCWB	4.9	4.4	6.4	9.6	13.0	13.4	15.9	14.0	13.5	10.5	8.5	7.1
	5%	DB	5.5	5.5	6.8	13.8	19.2	18.3	22.5	19.1	17.1	12.7	9.4	7.3
		MCWB	4.2	3.9	4.7	8.3	11.8	12.5	15.3	14.2	12.9	10.4	7.7	6.2
	10%	DB	4.6	4.6	5.7	11.2	16.6	16.5	20.2	17.8	15.7	11.2	8.3	6.3
		MCWB	3.3	3.2	4.1	6.1	10.8	11.7	14.8	13.2	12.2	8.7	6.7	5.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	5.8	8.6	13.0	15.4	16.3	18.2	16.7	16.2	13.2	10.2	8.6
		MCDB	6.8	6.9	10.9	16.7	22.2	20.7	24.7	20.0	20.9	15.1	11.6	8.9
	2%	WB	5.4	4.9	6.6	10.2	13.9	14.7	17.0	15.5	14.8	11.9	9.0	7.5
		MCDB	5.9	6.0	8.2	15.1	20.4	18.0	22.3	18.5	17.1	13.3	10.1	8.0
	5%	WB	4.6	4.2	5.5	8.4	12.5	13.4	16.1	14.6	13.7	10.6	8.1	6.3
		MCDB	5.4	5.4	6.5	12.8	17.2	17.1	20.7	17.9	16.3	12.3	9.2	7.1
	10%	WB	3.4	3.3	4.4	6.8	11.3	12.4	15.1	13.8	12.7	9.1	6.9	5.2
		MCDB	4.4	4.6	5.5	10.0	15.9	15.6	19.4	17.1	15.1	10.8	8.0	6.2
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.4	5.8	7.4	7.9	7.2	7.0	6.3	5.9	5.4	4.3	4.5
		MCDBR	4.6	5.5	7.6	11.9	12.4	10.9	11.5	8.1	8.6	7.5	5.2	4.4
		MCWBR	4.2	4.5	5.2	7.7	5.3	5.3	4.9	4.1	4.7	4.8	4.2	4.5
	5% WB	MCDBR	4.3	5.2	6.7	11.0	10.9	9.3	9.9	7.2	8.0	6.0	5.0	4.3
		MCWBR	4.3	4.5	5.2	8.0	5.4	5.5	4.7	4.1	4.9	4.6	4.5	4.5
(40) Clear-Sky Solar Irradiance	taub		0.266	0.296	0.320	0.354	0.350	0.344	0.358	0.356	0.335	0.317	0.280	0.245
	taud		2.171	2.271	2.291	2.294	2.430	2.477	2.453	2.455	2.511	2.469	2.281	2.103
	Ebn at Noon		542	724	817	849	882	892	868	843	805	705	531	424
	Edh at Noon		46	70	92	109	102	99	99	92	75	58	41	33
(44) All-Sky Solar Radiation	RadAvg		0.32	0.93	2.19	3.87	4.95	5.39	4.79	3.70	2.38	1.15	0.42	0.21
	RadStd		0.06	0.19	0.33	0.52	0.68	0.61	0.56	0.32	0.23	0.17	0.07	0.03

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-116	N/A	N/A

Nomenclature: See separate page